Stacking Galaxies Reveals a Universal Baryon Fountain

Averaging many simulated galaxies exposes a smooth, regular gas-cycling pattern hidden by individual diversity.

arXiv:2609.21374 — Stacked baryon cycles in cosmological hydrodynamic simulations

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Background

Galaxies continuously pull in gas and expel it through feedback, forming fountain-like cycles that shape how they grow. Individual systems are too diverse and dynamic to reveal the underlying pattern directly.

> **Observational hints of baryon cycles exist, but a concrete, unified picture of their geometry and physics remains elusive.**

What this paper is selling

The authors stack gas fields across ensembles of simulated galaxies, suppressing case-to-case noise. This unveils strikingly regular fountain patterns matching observations and yields a compact parameterized model.

Key results to remember

- Stacking recovers coherent inflow and outflow cones across the full galaxy ecosystem extent.
- A simple fountain model reproduces the stacked velocity field with small residuals.
- The same fountain geometry persists across stellar mass and cosmic time, scaling with halos.

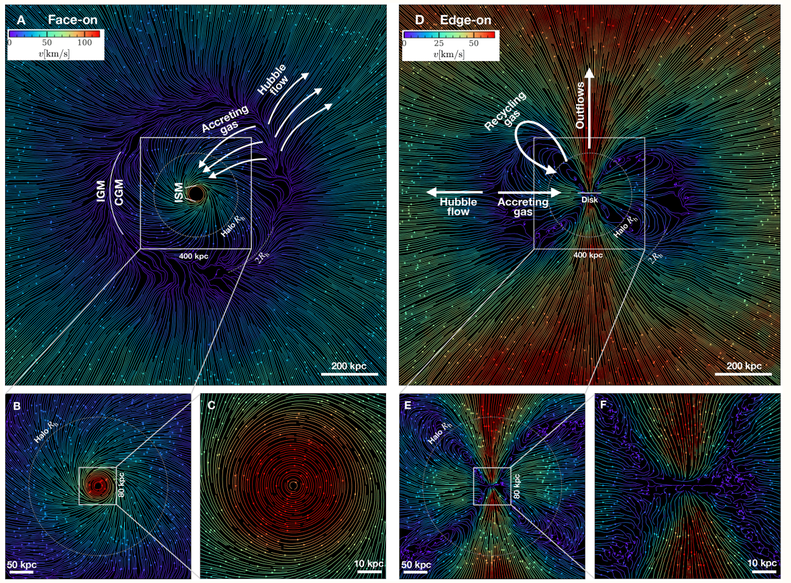


Figure 1 | Stacked velocity streamlines reveal a smooth, coherent fountain pattern invisible in any single galaxy.

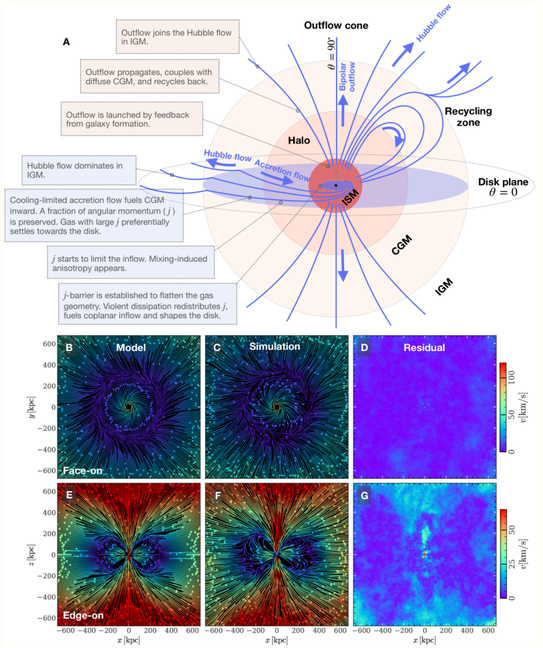


Figure 4 | A compact parameterized fountain model reproduces the simulated velocity field, compressing the physics into few numbers.